3.2 Incorporating what you already know into a Gaussian process

In this section, we will identify situations in which specifying prior knowledge in a GP is important. This discussion will motivate our discussions in the remaining portion of this chapter.

A prior GP may start out having a constant mean and CI everywhere. This GP then gets updated to smoothly interpolate the observed data points, as Figure 3.4 denotes. That is, the mean prediction exactly goes through the data points, and the 95% CI vanishes in those regions.

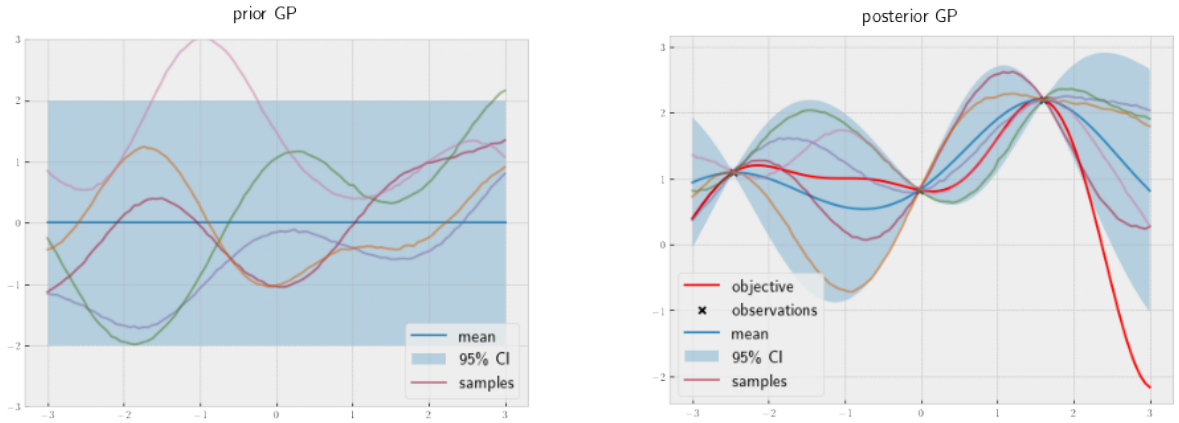


Figure 3.4 Comparison between a prior and a posterior GP. The prior GP contains prior information about the objective function, while the posterior GP combines that information with actual observations.

The prior GP in Figure 3.4 assumes nothing about the objective function we are modeling. That's why this GP's mean prediction is zero everywhere. In many cases, however, even though we don't know the exact form of our objective function, there are aspects of the objective that we *do* know.

For example:

- When modeling the accuracy of a model in a hyperparameter tuning application, we know that the range of the objective function is between 0 and 1.
- In our housing price example from Section 2.1, the function values (the prices) are strictly positive and should increase when a desirable property of the house, such as living area, increases.
- Again in the housing example, the function value is more sensitive to some features than others. For instance, the price of a house increases faster as a function of the number of stories than as a function of living area—one extra story increases the price of a house more than one extra squared foot of living area.

This kind of information is exactly the prior knowledge that we'd like to represent with a GP, and one of the biggest advantages of using a GP is that we have many ways to incorporate prior knowledge. Doing this helps close the gap between the GP surrogate and the actual objective

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